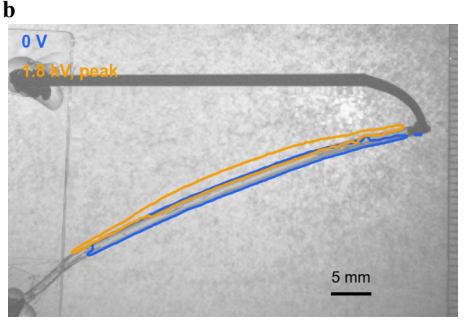
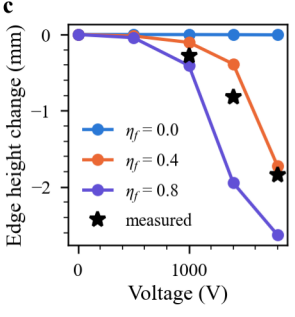
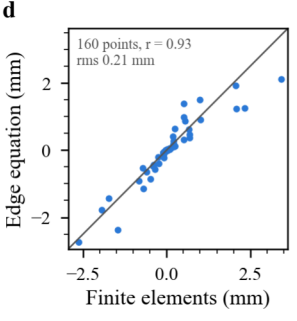
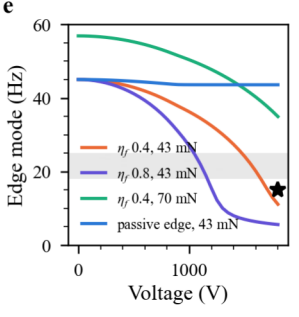
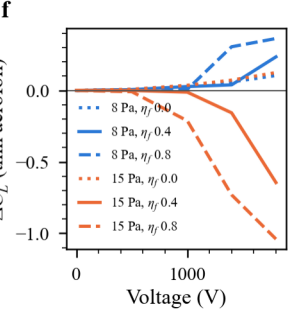

a
0 V
at voltage
edge tension T
load q
skin N/c
edge slack: T ≈ 0
load q
skin N/c
Δz = q / (8T/L² + N/c)
edge mode
45 Hz
11–27 Hz


b
0 V
1.6 kV, peak
5 mm


c
Edge height change (mm)
0
−1
−2
0
1000
Voltage (V)
$\eta_f = 0.0$
$\eta_f = 0.4$
$\eta_f = 0.8$
measured


d
160 points, r = 0.93
rms 0.21 mm
Edge equation (mm)
2
0
−2
−2.5
0.0
2.5
Finite elements (mm)


e
Edge mode (Hz)
60
40
20
0
0
1000
Voltage (V)
η_f 0.4, 43 mN
η_f 0.8, 43 mN
η_f 0.4, 70 mN
passive edge, 43 mN


f
ΔC_L (thin aerofoil)
0.0
−0.5
−1.0
0
1000
Voltage (V)
8 Pa, η_f 0.0
8 Pa, η_f 0.4
8 Pa, η_f 0.8
15 Pa, η_f 0.0
15 Pa, η_f 0.4
15 Pa, η_f 0.8